

communication between future CMOS circuits.

As shown in Fig. 3, first the electrical signal is converted into an optical signal by using an optical modulator imposed by laser or any optoelectronics device like LED or photodiode. The signal propagates through the optical cable using waveguide. At the receiver side, the photodetector receives the optical signal and converts it into an electrical signal and amplifies.

By using optics many physical problems related to interconnects can be solved. Optics may solve design problems such as crosstalk noise, bandwidth, isolation and impedance matching, and also reduce problems associated with long-distance communication.

Need for optical system

Two important parts of an on-chip optical interconnect are optical devices

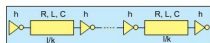


Fig. 2(a): General structure of interconnects

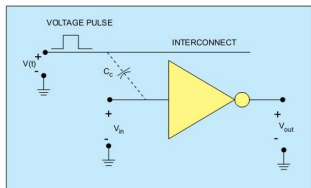


Fig. 2(b): Effect of coupling capacitor between two interconnect lines

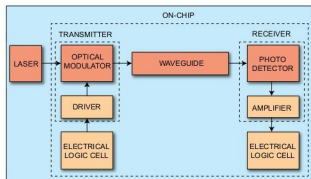


Fig. 3: Block diagram of optical interconnect

called 'optoelectronics devices' and optical system called 'medium.' Here we discuss some basic requirements of an optical system for on-chip optical interconnects. There are two broad categories of the optical system: 'guided-wave' optical system and 'free space' optical system.

For guided-wave optical system, one can use a thin optical fibre cable, waveguide etc. An optical fibre having a diameter of up to $10\text{ }\mu\text{m}$ can be used for on-chip optical interconnect. It has many advantages over other devices such as:

1. No crosstalk inside the optical fibre cable
2. Higher bandwidth, therefore it can operate at a higher data rate
3. Reduced loss due to signal attenuation
4. Small size and light-weight

In waveguide approach, one has to use a waveguide having a diameter of up to $10\text{ }\mu\text{m}$ for on-chip optical interconnect system. Smaller optical guides of the order of micrometre size should have low enough loss so that the propagation loss does not become

a big issue for the chip. So for simplicity we connect the waveguide from the central area to the edges. Whether such guides could handle the internal on-chip interconnects strongly depends on the architecture.

The other approach is to use a 'free space' system where multiple light beams can go in and out of the chips, usually in the form of arrays. So arrays of beams go into and out of the chip. Free-space approach can easily handle a large number of arrays of beams. So for high resolution and focusing on the

scale of micrometre size, we use an ordinary lens. Optical elements are fabricated using lithographic patterning. These can generate very large regular arrays of spots from a single laser beam and implement a variety of quite complex regular interconnection patterns.

Requirement for optoelectronic devices

As shown in Fig. 3, an on-chip optical interconnect system consists of transmitter driver circuit, optical channel and receiver circuit. Here p-i-n photodiodes and metal-semiconductor metal (MSM) photodetectors work as the receiver circuit and vertical-cavity surface-emitting laser (VCSEL) and quantum-well modulator work as the transmitter for optical output. The photodetector in the receiver part converts optical energy into electrical energy. In the next step, this electrical signal is amplified by the amplifier circuit. So MSM photodetector gives a fast response and excellent quantum efficiency.

Optical output devices and transmitter circuits. Quantum-well modulator and VCSEL are important devices for the transmitter of on-chip optical interconnect system. You can also use an LED as the transmitter. It will make the fabrication process easier than with VCSELs and avoid some problems of lasers. We discuss here quantum-well modulator and VCSELs in short.

Quantum-well modulator. These modulators are used in demonstrating actual dense interconnects to and from silicon CMOS chips. They are made in large arrays and require an external beam. This requirement can be considered as an advantage or disadvantage of this system. The external beam is required to generate an array and to separate the incident and reflected beams. So we draw a setup that can easily handle the one-master laser beam. The single-master laser beam can be diffracted into arrays of equal beams by using a diffractive beam generator. Additional master laser beam allows centralised clocking of the entire system. Modulators avoid many of the problems of mode quality,